Use Case Testing
(1400 – 1600)

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Use Case Testing

- A use case specifies the behavior of a system that interacts with one or more actors resulting in an observable result of value to the actors.
- The use case specification should consist of
  - Preconditions (optional)
  - Post-conditions (observable results) [success guarantees]
  - Only one main path (sometimes called the happy path) [main success scenario]
  - Alternate paths (0 or more) [technology & data variation list]
  - Exception paths (0 or more) [extensions]

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- **T**he coverage is usually measured by the number of use case paths tested divided by the total number of use case paths.
- By **"**paths**",** means all possible sequences of actions that go through all possible exceptions and alternate paths.
- Hence, to cover the use case, we need
  - One test case to cover the main scenario
  - One separate test case for each exception
  - One separate test case for each alternative

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Use Case	Pay Fee
Use Case ID	UC006
Actors	Parents
Pre-conditions	User has valid parents credentials There is an active network connection
Normal Flow	1. The use case begins when parents select "Payment Logs" 2. System displays the overall information of the payment, paid fees with payment date and checker column 3. On the bottom, total paid amount and unpaid amount is displayed 4. User select fees to pay at checker column 5. User select "Pay Now" 6. System display popup modal with fees and amount confirmation 7. User select confirm 8. User select payment methods. 9. If user select "Close" from popup modal, Exception 1 is performed 10. If the transaction is successful, Alternative 1 is performed 11. If the transaction is unsuccessful, Alternative 2 is performed 12. The use case ends
Alternative Flow	**1. Successful Transaction** 1.1. System display success transaction message and change the payment status to "Paid" with payment date 1.2. The use case resumes at Normal Flow 7 **2. Unsuccessful Transaction** 2.1. System display unsuccess transaction message 2.2. The use case resumes at Normal flow 2
Exception	**1. Cancel Payment** 1.1. System close popup modal 1.2. The use case resumes at Normal Flow 2
Related Requirement	
Post-conditions	User able to pay fee, payment date, total and unpaid fees is updated

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Use Case Testing

Test Case #	Path covered
1	Normal flow: 1 - 12
2	Alternative 1: Normal Flow 1 -10, Alternative Flow 1
3	Alternative 2: Normal Flow 1 -11, Alternative Flow 2
4	Exception 1: Normal Flow 1 - 9, Exception Flow 1

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Use Case Testing: Test Case #1

TC1: Precondition:
1) User has valid credentials
2) There is an active network

#	Test Steps	Expected Results
1	The use case begins when parents select "Payment Logs"	Payment logs able to be selected
2	System displays the overall information of the payment, paid fees with payment date and checker column	The payment, paid fees with payment date and checker column is displayed
3	On the bottom, total paid amount and unpaid amount is displayed	Total paid amount and unpaid amount is displayed at bottom page
4	User select fees to pay at checker column Test data: Fee: 80.00	
5	User select "Pay Now"	Pay Now is able to be selected
6	System display popup modal with fees and amount confirmation : :	The fee is updated

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Use Case Testing: Test Case #2

TC1: Precondition:
1) User has valid credentials
2) There is an active network

#	Test Steps	Expected Results
1	The use case begins when parents select "Payment Logs"	Payment logs able to be selected
2	System displays the overall information of the payment, paid fees with payment date and checker column	The payment, paid fees with payment date and checker column is displayed
3	On the bottom, total paid amount and unpaid amount is displayed	Total paid amount and unpaid amount is displayed at bottom page
4	User select fees to pay at checker column Test data: Fee: 80.00	
5	:	
10	System display success transaction message and change the payment status to "Paid" with payment date	The fee is updated. Status change to Paid